**Back Substitution Method**

**Question 1**

$T(n) = 2T(n-1) + 1$
$T(0) = 1$

$T(n-1) = 2T(n-2) + 1$
$T(n) = 2T(n-1) + 1$
$T(n) = 2(2T(n-2) + 1) + 1$
**$T(n) = 4T(n-2) + 2 + 1$**

$T(n-2) = 2T(n-3) + 1$
$T(n) = 4T(n-2) + 2 + 1$
$T(n) = 4T(2T(n-3) + 1) + 2 + 1$
**$T(n) = 8T(n-3) + 4 + 2 + 1$**

$T(n-3) = 2T(n-4) + 1$
$T(n) = 8T(n-3) + 4 + 2 + 1$
$T(n) = 8T(2T(n-4) + 1) + 4 + 2 + 1$
**$T(n) = 16T(n-4) + 8 + 4 + 2 + 1$**

$T(n-4) = 2T(n-5) + 1$
$T(n) = 16T(n-4) + 8 + 4 + 2 + 1$
$T(n) = 16T(2T(n-5) + 1) + 8 + 4 + 2 + 1$
**$T(n) = 32T(n-5) + 16 + 8 + 4 + 2 + 1$**

$1 = 2^0$
$2 = 2^1$
$4 = 2^2$
$8 = 2^3$
$16 = 2^4$
Answer: **$T(n) = O(2^n)$**

**Question 2**

$T(n) = T(n-2) + n^2$
$T(0) = 1$
n=2k so we have to go up by 2s not 1s

$T(n-2) = T(n-4) + (n-2)^2$
$T(n) = T(n-2) + n^2$
$T(n) = T(T(n-4) + (n-2)^2) + n^2$
**$T(n) = T(n-4) + (n-2)^2 + n^2$**

$T(n-4) = T(n-6) + (n-4)^2$
$T(n) = T(n-4) + (n-2)^2 + n^2$
$T(n) = T(n-6) + (n-4)^2 + (n-2)^2 + n^2$

$T(n-6) = T(n-8) + (n-6)^2$
$T(n) = T(n-6) + (n-4)^2 + (n-2)^2 + n^2$
$T(n) = T(n-8 + (n-6)^2 + (n-4)^2 + (n-2)^2 + n^2$

$K = n/2$
$T(n) = n(n + 1)(2n + 1) / 6$
$T(n) = 1 + 4 * (n/2)(n/2 + 1)(2*(n/2) + 1) / 6$
$T(n) = 1 + n(n+1)(n+2) / 6$
$T(n) = 1 + (n^3 + 3n^2 + 2n) / 6$
Answer: **$T(n) = O(n^3)$**

**Question 3**

$T(n) = T(n-1) + 1/n$
$T(1) = 1$

$T(n) = T(n-1) + 1/n$
$T(n) = T(n-2) + 1/(n-1) + 1/n$
$T(n) = T(n-3) + 1/(n-2) + 1/(n-1) + 1/n$
$T(n) = T(1) + 1/2 + 1/3 + ... + 1/n$
$T(n) = 1 + 1/2 + 1/3 + ... + 1/n$
harmonic series from web: $H_n \approx \ln(n) + \gamma$
$T(n) = H_n \approx \ln(n) + \gamma$ we drop the constant and that give us the runtime
$T(n) = \ln(n)$
Answer: **$T(n) = O(\log(n))$**

**Master Theorem:**

**Question 4**
$T(n) = 2T(n/4) + 1$
$T(0) = 1$

$a = 2$
$b = 4$
$f(n) = n^0$
$\log_b(a) = \log(2) / \log(4) = 0.5$
**f(n) is less than 0.5**
Answer: $T(n) =$ **$O(n^{0.5})$**

**Question 5**
$T(n) = 2T(n/4) = n^{1/2}$
$T(0) = 1$

$a = 2$
$b = 4$
$f(n) = n^{1/2}$
$\log_b(a) = \log(2) / \log(4) = 0.5$
**f(n) is equal to 0.5 so both are valid**
Answer: $T(n) =$ **$O(n^{1/2})$**

**Question 6**
$T(n) = 2T(n/4) + n^2$
$T(0) = 1$

$a = 2$
$b = 4$
$f(n) = n^2$
$\log_b(a) = \log_4(2)$
$\log_b(a) = \log(2) / \log(4) = 0.5$
$\log_b(a) = 0.5$
**f(n) is worse than 0.5**
Answer: $T(n) =$ **$O(n^2)$**

**Question 7**
$T(n) = 10T(n/3) + n^2$
$T(0) = 1$

$a = 10$
$b = 3$
$f(n) = n^2$
$\log_b(a) = \log_3(10)$
$\log_b(a) = \log(10) / \log(3) = 2.095$
**f(n) is less than 2.095**
Answer: $T(n) =$ **$O(n^{2.10})$**

**Question 8**
$T(n) = 2T(2n/3) + 1$
$T(0) = 1$

$a = 2$
$b = 3/2$
$c = 1$

$\log_b(a) = \log_{(3/2)}(2)$
$\log_b(a) = \log(2) / \log(3/2) = 1.7095$
**f(n) is less than 1.7095**
Answer: $T(n) =$ **$O(n^{1.71})$**